

Covered: Construction of $\mathcal{L}^p$ space and its norm, Minkowski and Hölder Inequalities.

1 L^p Spaces

$(\Omega, \mathcal{F}, \mu)$ a measure space. Fix $1 \leq p < \infty$. Define

$$L^p := \left\{ f : \Omega \rightarrow \overline{\mathbb{R}} \mid f \text{ measurable and } \int_{\Omega} |f|^p \, d\mu < \infty \right\}.$$

Remark 1.1. L^1 is a vector space over $\mathbb{R}$. Since if $p = 1$, then for $f, g \in L^1$ and $\alpha, \beta \in \mathbb{R}$,

$$|\alpha f + \beta g| \leq |\alpha| |f| + |\beta| |g| \implies \alpha f + \beta g \in L^1$$

Also, if $f \in L^1$, then

$$\mu(\{x \in \Omega : |f(x)| = \infty\}) = 0,$$

so we may, up to a null set, regard f as a map $f : \Omega \rightarrow \mathbb{R}$

The following inequalities will be useful

Theorem 1.2 (Hölder's inequality). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ be measurable. If $p, q > 1$ and $\frac{1}{p} + \frac{1}{q} = 1$, then

$$\int_{\Omega} |fg| \, d\mu \leq \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} \left(\int_{\Omega} |g|^q \, d\mu \right)^{1/q}.$$

Corollary 1.3 (Cauchy–Schwarz inequality). Taking $p = q = 2$,

$$\int_{\Omega} |fg| \, d\mu \leq \left(\int_{\Omega} |f|^2 \, d\mu \right)^{1/2} \left(\int_{\Omega} |g|^2 \, d\mu \right)^{1/2}.$$

Theorem 1.4 (Minkowski's inequality). Let $f, g : \Omega \rightarrow \overline{\mathbb{R}}$ be measurable and $1 \leq p < \infty$. Define

$$\|f\|_p := \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p}.$$

Then

$$\|f + g\|_p \leq \|f\|_p + \|g\|_p.$$

Note. Over a space X , the norm is a function $\|\cdot\| : X \rightarrow \mathbb{R}$ such that for $x, y \in X$ and $\alpha \in \mathbb{R}$,

1. Positivity: $\|x\| \geq 0$, $\|x\| = 0 \iff x = 0$
2. Absolute homogeneity: $\|\alpha x\| = |\alpha| \|x\|$
3. Triangle inequality: $\|x + y\| \leq \|x\| + \|y\|$

Define,

$$\|f\|_1 := \int_{\Omega} |f| \, d\mu.$$

The above function satisfies the first two conditions for norm naturally,

$$\|\alpha f\|_1 = |\alpha| \|f\|_1, \quad \|f + g\|_1 \leq \|f\|_1 + \|g\|_1$$

However the above norm can be 0 at two different functions as long as it is almost every where zero. Consequently, the function $\|\cdot\|_1$ only becomes a genuine norm after identifying functions which agree almost everywhere.

$$\|f\|_1 = 0 \iff \int_{\Omega} |f| \, d\mu = 0 \iff f = 0 \text{ a.e.}$$

Fix $1 \leq p < \infty$. Then

$$L^p = \left\{ f : \Omega \rightarrow \mathbb{R} \mid f \text{ measurable and } \int_{\Omega} |f|^p \, d\mu < \infty \right\},$$

and define

$$\|\cdot\|_p : L^p \rightarrow \mathbb{R}, \quad f \mapsto \|f\|_p = \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p}.$$

Then:

$$\begin{aligned} \|\alpha f\|_p &= |\alpha| \|f\|_p, \\ \|f + g\|_p &\leq \|f\|_p + \|g\|_p, \\ \|f\|_p &= 0 \iff f = 0 \text{ a.e.} \end{aligned}$$

Define a relation $\sim$ on L^p by

$$f \sim g \iff f - g = 0 \text{ a.e. } [\mu].$$

We claim that $\sim$ is an equivalence relation. Reflexivity and symmetry is obvious. Transitivity holds since.

$$\begin{aligned} f \sim g &\implies f - g = 0 \text{ a.e.} \\ g \sim h &\implies g - h = 0 \text{ a.e.} \end{aligned}$$

and therefore

$$f - h = (f - g) + (g - h) = 0 \text{ a.e.,}$$

since the union of two null sets is a null set.

Definition 1.5. We define the space,

$$\mathcal{L}^p := L^p / \sim = \{[f] \mid f \in L^p\},$$

where the candidate elements are $[f] = \{g \in L^p \mid g \sim f\}$.

The norm is well-defined, since

$$\begin{aligned} f \sim g &\implies f = g \text{ a.e.} \implies |f|^p = |g|^p \text{ a.e.} \\ &\implies \int_{\Omega} |f|^p \, d\mu = \int_{\Omega} |g|^p \, d\mu \implies \|f\|_p = \|g\|_p \end{aligned}$$

We now have all the requirements for a sound Lebesgue integrable function space, equipped with a norm.

Theorem 1.6. Let $(\Omega, \mathcal{F}, \mu)$ be a measure space and let $1 \leq p < \infty$. Then

1. $\mathcal{L}^p$ is a linear space;
 2. $\|\cdot\|_p$ is a norm on $\mathcal{L}^p$.
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Proof. (i) We define addition and scalar multiplication on $\mathcal{L}^p$ by

$$\begin{aligned}\mathcal{L}^p \times \mathcal{L}^p &\rightarrow \mathcal{L}^p, & ([f], [g]) &\mapsto [f + g] \\ \mathbb{R} \times \mathcal{L}^p &\rightarrow \mathcal{L}^p, & \alpha, [f] &\mapsto [\alpha f]\end{aligned}$$

These operations are well-defined, and the vector space axioms follow from the corresponding pointwise operations on representatives. Thus $(\mathcal{L}^p, +, \cdot)$ is a linear space.

(ii) We verify the norm axioms. First,

$$\begin{aligned}\|\alpha[f]\|_p &= \|[\alpha f]\|_p = \left(\int_{\Omega} |\alpha f|^p \, d\mu \right)^{1/p} = |\alpha| \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} \\ &\implies \boxed{\|\alpha[f]\|_p = |\alpha| \| [f] \|_p}\end{aligned}$$

By Minkowski's inequality,

$$\begin{aligned}\|[f] + [g]\|_p &= \|[f + g]\|_p \\ &= \left(\int_{\Omega} |f + g|^p \, d\mu \right)^{1/p} \\ &\leq \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p} + \left(\int_{\Omega} |g|^p \, d\mu \right)^{1/p} = \|[f]\|_p + \|[g]\|_p.\end{aligned}$$

Finally,

$$\|[f]\|_p = 0 \iff \|f\|_p = 0 \iff f = 0 \text{ a.e. } [\mu] \iff [f] = [0].$$

Hence $\|\cdot\|_p$ is a norm on $\mathcal{L}^p$. □

Before proceeding with the proof of Minkowski's Inequality, we have the following lemma.

Lemma 1.7. For $a, b \geq 0$, we have $(a + b)^p \leq 2^{p-1}(a^p + b^p)$

Proof. Since $t \mapsto t^p$ is convex on $[0, \infty)$,

$$\left(\frac{a + b}{2} \right)^p \leq \frac{a^p + b^p}{2}.$$

Multiplying by 2^p gives

$$(a + b)^p \leq 2^{p-1}(a^p + b^p).$$
□

Theorem 1.8 (Minkowski's Inequality). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, $f, g : \Omega \rightarrow \mathbb{R}$ measurable, and $1 \leq p < \infty$. Define

$$\|f\|_p := \left(\int_{\Omega} |f|^p \, d\mu \right)^{1/p}.$$

Then

$$\boxed{\|f + g\|_p \leq \|f\|_p + \|g\|_p}.$$

Proof. Step 1. Without loss of generality, assume $\|f\|_p$ and $\|g\|_p$ are finite, otherwise the inequality is immediate.

Step 2. We first show that $\|f + g\|_p$ is finite. It is enough to show that

$$\int_{\Omega} |f + g|^p \, d\mu < \infty.$$

Using the above lemma together with triangle inequality,

$$\begin{aligned} |f + g| &\leq |f| + |g| \quad \text{pointwise} \\ \implies |f + g|^p &\leq (|f| + |g|)^p \leq 2^{p-1}(|f|^p + |g|^p) \\ \implies \int_{\Omega} |f + g|^p \, d\mu &\leq 2^{p-1} \left(\int_{\Omega} |f|^p \, d\mu + \int_{\Omega} |g|^p \, d\mu \right) < \infty \end{aligned}$$

Thus $f + g \in L^p$

Step 3. We now prove $\|f + g\|_p \leq \|f\|_p + \|g\|_p$. Since

$$\|f + g\|_p^p = \int_{\Omega} |f + g|^p \, d\mu, \quad p \geq 1,$$

it suffices to establish the corresponding integral inequality.

Without loss of generality, we may assume $f, g \geq 0$ with $\|f\|_p$ and $\|g\|_p$ finite, we have

$$\begin{aligned} \|f + g\|_p^p &= \int_{\Omega} (f + g)^p \, d\mu \\ &= \int_{\Omega} (f + g)^{p-1} f \, d\mu + \int_{\Omega} (f + g)^{p-1} g \, d\mu. \end{aligned}$$

By Hölder's inequality,

$$\begin{aligned} \int_{\Omega} (f + g)^{p-1} f \, d\mu &\leq \left(\int_{\Omega} f^p \, d\mu \right)^{1/p} \left(\int_{\Omega} (f + g)^p \, d\mu \right)^{(p-1)/p} \\ &= \|f\|_p \|f + g\|_p^{p-1}, \end{aligned}$$

and similarly,

$$\int_{\Omega} (f + g)^{p-1} g \, d\mu \leq \|g\|_p \|f + g\|_p^{p-1}.$$

Therefore

$$\|f + g\|_p^p \leq (\|f\|_p + \|g\|_p) \|f + g\|_p^{p-1}.$$

If $\|f + g\|_p = 0$, the result is immediate. Otherwise, dividing by $\|f + g\|_p^{p-1}$,

$$\boxed{\|f + g\|_p \leq \|f\|_p + \|g\|_p}.$$

For arbitrary f, g , since

$$|f + g| \leq |f| + |g|,$$

we may replace f, g by $|f|, |g|$, obtaining the general case. $\square$

Lemma 1.9 (Young's Inequality). Let $p, q > 1$ satisfy $\frac{1}{p} + \frac{1}{q} = 1$. Then, for every $u, v \geq 0$,

$$\boxed{uv \leq \frac{u^p}{p} + \frac{v^q}{q}}.$$

Proof. For $u, v > 0$, write $u = e^{s/p}$ and $v = e^{t/q}$ for some. Since $1/p + 1/q = 1$, convexity of e^x gives

$$uv = e^{s/p+t/q} \leq \frac{1}{p}e^s + \frac{1}{q}e^t = \frac{u^p}{p} + \frac{v^q}{q}.$$

The cases $u = 0$ or $v = 0$ are immediate. $\square$

We can now move on to the proof of Hölder's Inequality

Theorem 1.10 (Hölder's Inequality). Let $(\Omega, \mathcal{F}, \mu)$ be a measure space, let $f, g : \Omega \rightarrow \mathbb{R}$ be measurable, and let

$$p, q > 1, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$\int_{\Omega} |fg| \, d\mu \leq \|f\|_p \|g\|_q.$$

Proof. Without loss of generality, we may assume $f, g \geq 0$, since $|fg| = |f| |g|$.

Set $A := \|f\|_p$ and $B := \|g\|_q$.

If $A = \infty$ or $B = \infty$, the inequality is immediate. Thus assume both finite.

If $A = 0$ or $B = 0$, then $f = 0$ a.e. or $g = 0$ a.e., and the result is immediate. Hence assume they are strictly greater than 0.

It remains to show

$$\int_{\Omega} \frac{fg}{AB} \, d\mu \leq 1.$$

Applying Young's inequality pointwise with $u = \frac{f(\omega)}{A}$ and $v = \frac{g(\omega)}{B}$, we obtain

$$\frac{f(\omega)}{A} \frac{g(\omega)}{B} \leq \frac{1}{p} \frac{f(\omega)^p}{A^p} + \frac{1}{q} \frac{g(\omega)^q}{B^q}.$$

Integrating,

$$\int_{\Omega} \frac{fg}{AB} \, d\mu \leq \frac{1}{pA^p} \int_{\Omega} f^p \, d\mu + \frac{1}{qB^q} \int_{\Omega} g^q \, d\mu = \frac{1}{p} + \frac{1}{q} = 1.$$

Therefore

$$\int_{\Omega} fg \, d\mu \leq AB = \|f\|_p \|g\|_q.$$

□